

Basic Principles of Medicine 2

Nervous System and Behavioral Sciences

IMCQ 1

St. George's University

1. A 25-year-old woman at 18 weeks' gestation comes for a routine ultrasound examination. Ultrasonography shows a fetus with a neural tube defect characterized by a protrusion of spinal cord tissues through an opening in the vertebral column. Which of the following developmental processes is most likely associated with this condition?

- A. Myelination of the spinal cord
- B. Neural crest migration
- C. Primary neurulation
- D. Secondary neurulation

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2. During postnatal examination, a newborn is observed to have a tuft of hair over the midline lumbosacral region. Which of the following pathological processes best explains the findings in this patient?

- A. Ageneration of the cerebellar vermis
- B. Failure of closure of the anterior neuropore
- C. Cavitation in the spinal canal
- D. Failure of closure of the posterior neuropore
- E. Tethering of the spinal cord to the vertebrae

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3. Investigators are studying the functional properties of the spinal cord. They observe a transition from somatosensory function to motor function, corresponding to a landmark that runs longitudinally. Which of the following sulci most likely corresponds to the landmark of interest?

- A. Calcarine
- B. Central
- C. Limitans
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4. A 60-year-old man comes to the physician because of a 1-day history of drooling and difficulty speaking. Neurologic examination shows weakness of the facial muscles on the left side of his face. Which of the following structures most likely gives rise to the affected cranial nerve?

- A. Telencephalon
- B. Mesencephalon
- C. Myelon
- D. Metencephalon
- E. Diencephalon

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5. A 70-year-old man is brought to the emergency department because of a 2-hour history of double vision. Neurologic examination shows dysfunction of a nerve that exits the ventral surface of the midbrain. Which of the following nerves is most likely affected in this patient?

- A. Abducens
- B. Facial
- C. Oculomotor
- D. Trigeminal
- E. Vestibulocochlear

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6. A 48-year-old man comes to the physician because of a 1-week history of headaches. His vital signs are within normal limits. MRI of the brain shows an infarction of the postcentral gyrus in the left parietal lobe. Which of the following outcomes is most likely in this patient?

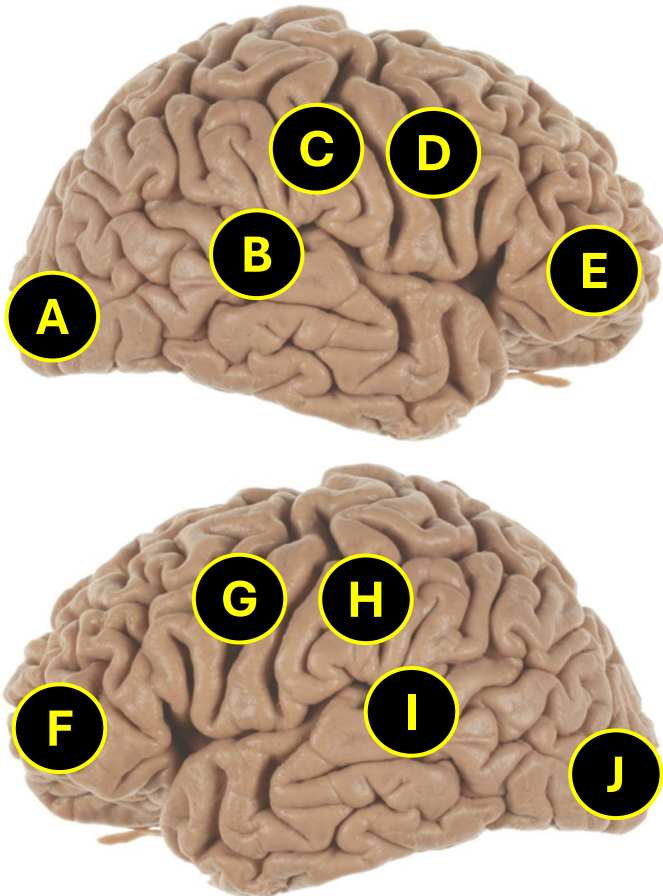
- A. Deafness
- B. Blindness
- C. Numbness
- D. Weakness
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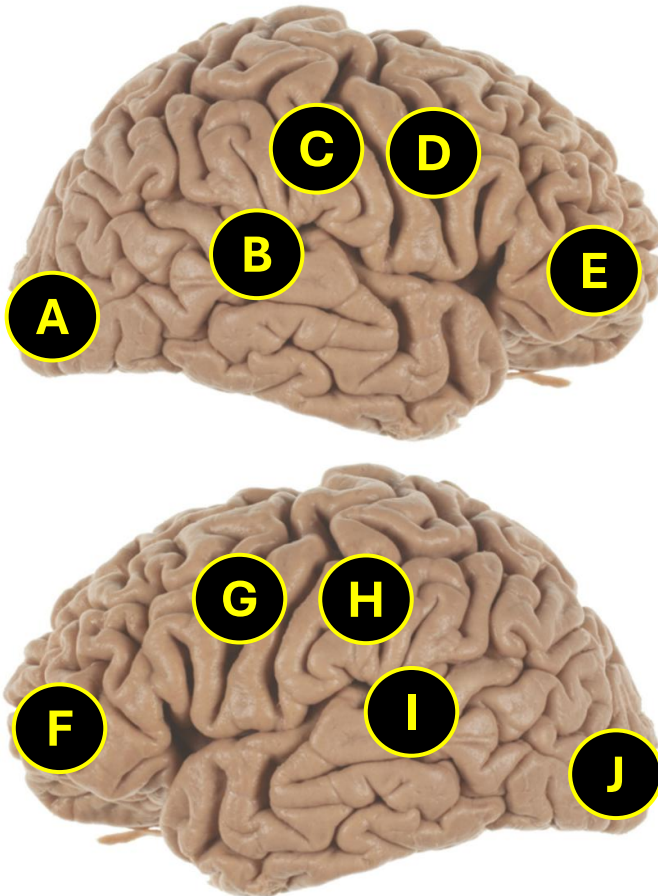
7. A 73-year-old woman comes to the physician because of a 6-hour history of difficulty with her vision. Neurologic examination shows blindness to objects located to her right. MRI of the brain shows an infarction affecting the left cerebral cortex. Which of the following labeled locations in the normal brain specimens shown is most likely damaged in this patient?

- A. A
- B. B
- C. C
- D. D
- E. E
- F. F
- G. G
- H. H
- I. I
- J. J



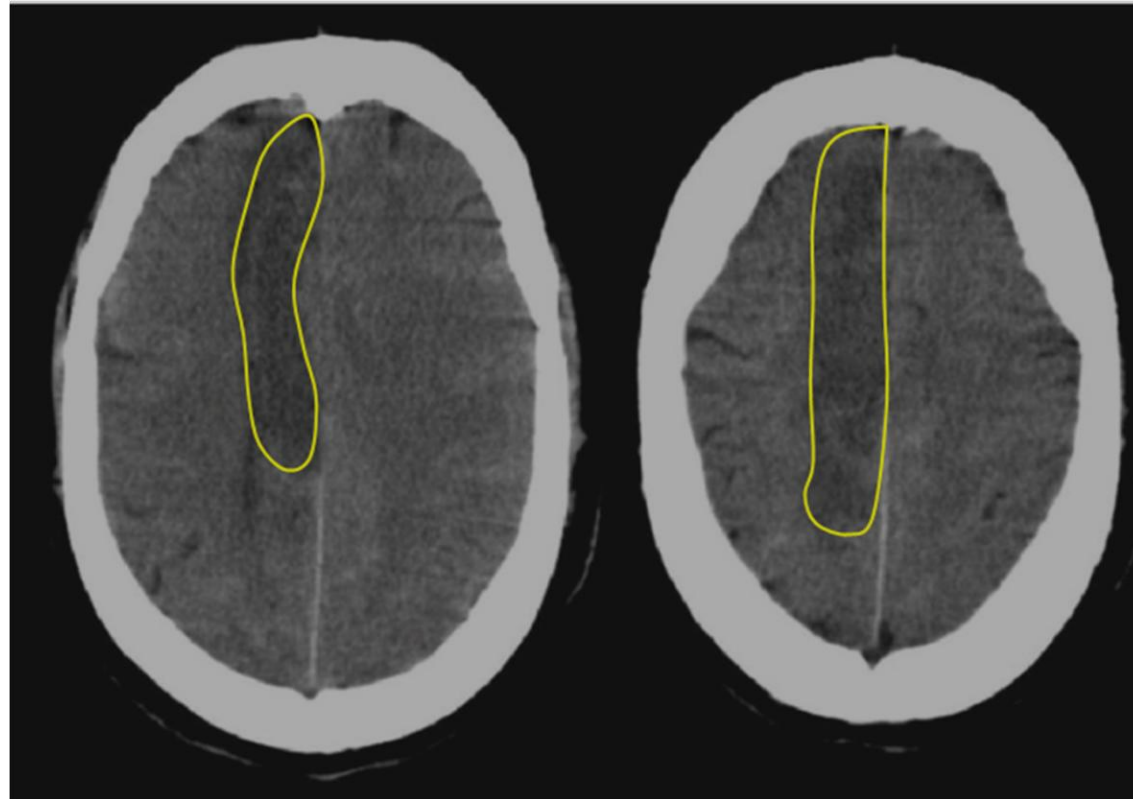
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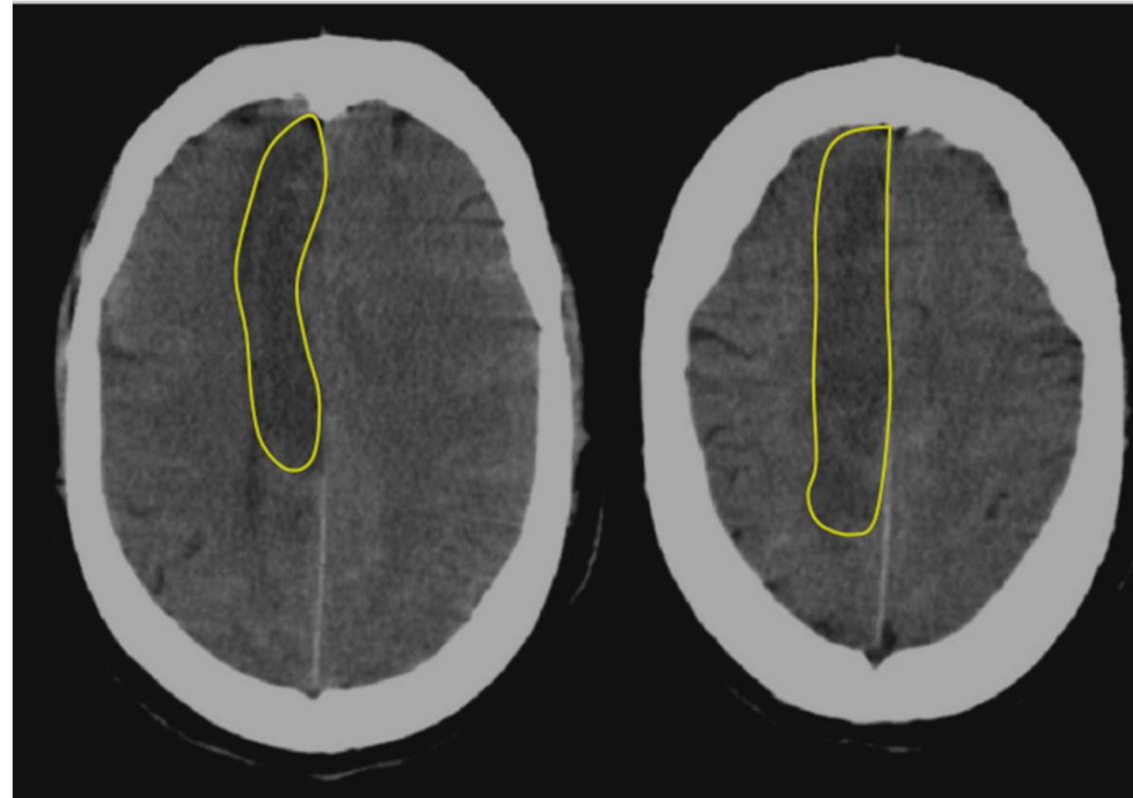
8. A 58-year-old man is brought to the emergency department because of a 2-day history of unilateral weakness. Neuroimaging studies of the brain show an infarction (see areas encircled in yellow in the figure below). Which of the following combinations best describes the imaging technique, the weakened side of the body, and the occluded cerebral artery?

- A. CT; left; anterior
- B. CT; left; middle
- C. CT; left; posterior
- D. CT; right; anterior
- E. CT; right; middle
- F. CT; right; posterior
- G. MRI; left; anterior
- H. MRI; left; middle
- I. MRI; left; posterior
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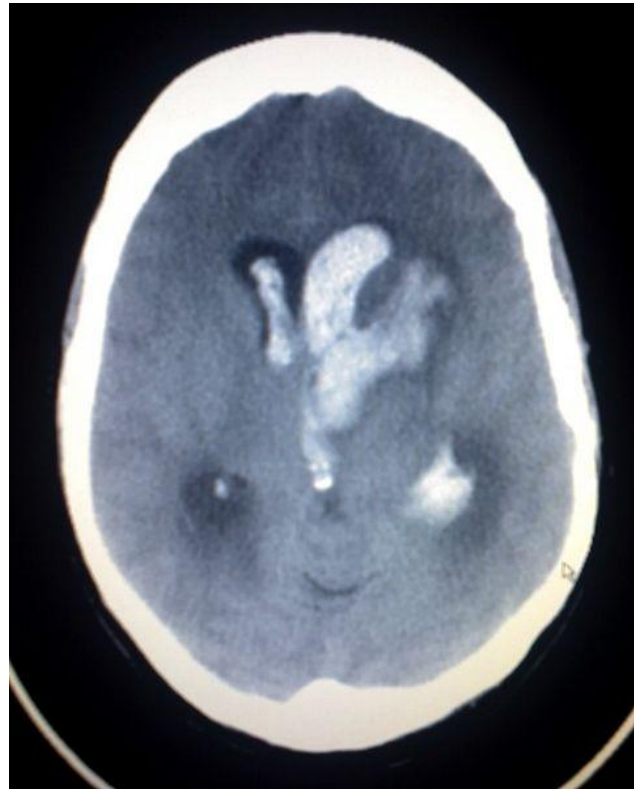
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SOM.MK.I.BPM2.1.NB.1.NSCI.0072
Identify the major intracranial
subarachnoid vessels and their general
cerebral distributions
SOM.CS.I.BPM2.1.NB.1.NSCI.0049
Describe the physics, diagnostic use and
contraindications / limitations of CT

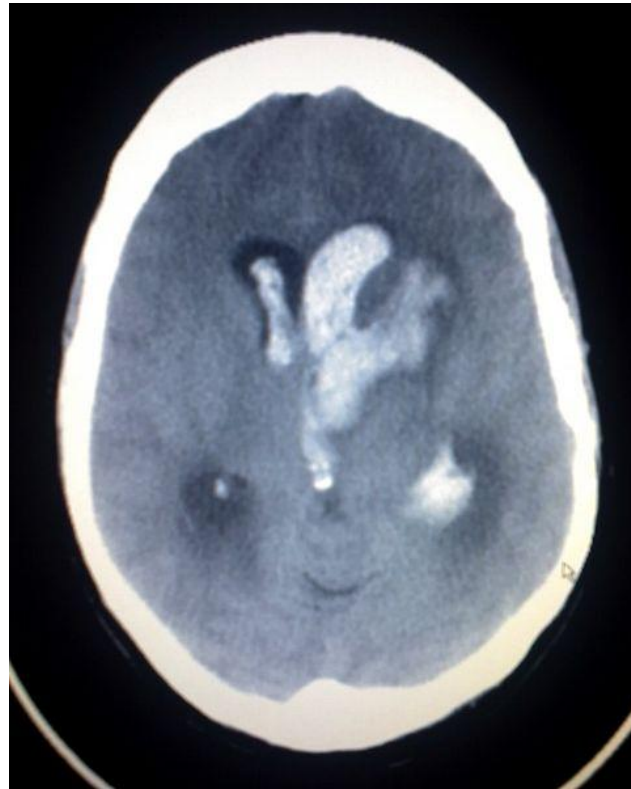
9. A 71-year-old man is brought to the emergency department because of a 12-hour history of weakness and intermittent unconsciousness. Imaging studies show a hemorrhage. Which of the following combinations best applies to the hemorrhage type and the affected cerebral artery in this patient?

- A. Epidural; middle
- B. Subdural; middle
- C. Subarachnoid; middle
- D. Intracerebral; middle
- E. Epidural; anterior
- F. Subdural; anterior
- G. Subarachnoid; anterior
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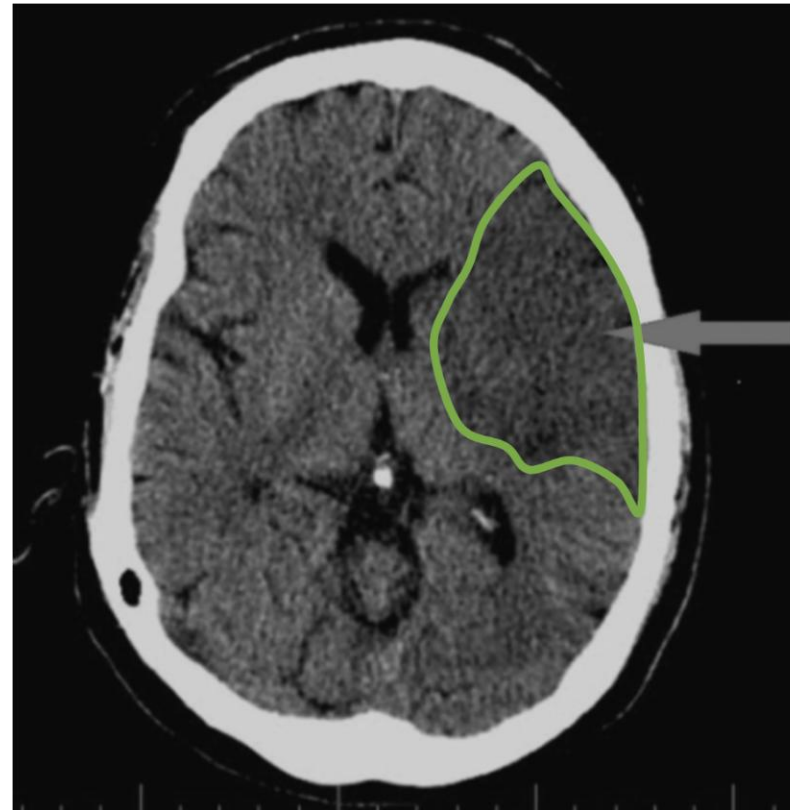
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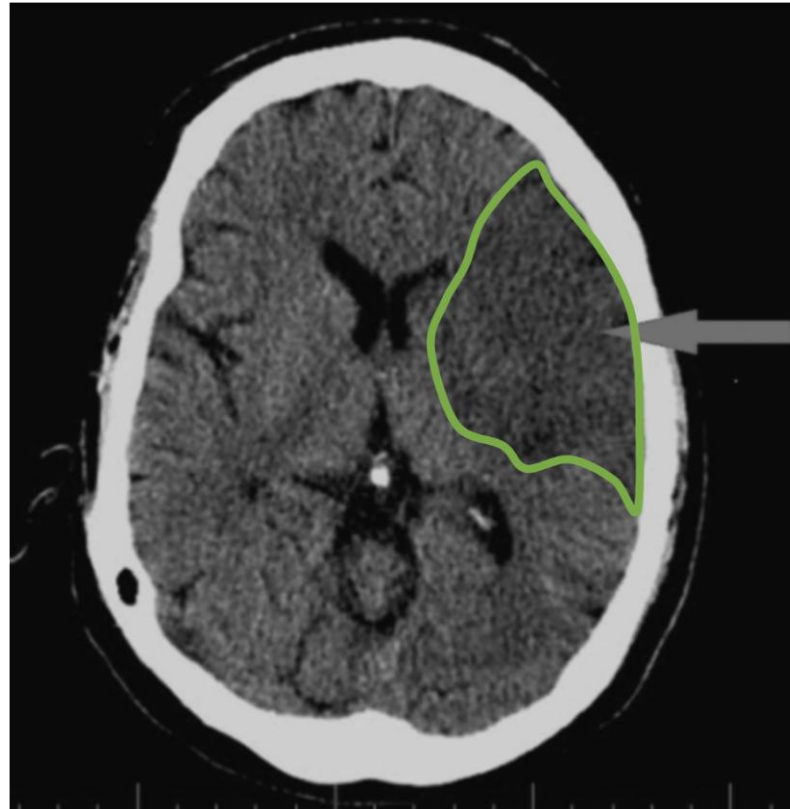
10. A 63-year-old woman is brought to the emergency department because of a 24-hour history of weakness and a loss of sensations. Neurologic examination shows unilateral weakness and abnormal reflexes. Somatosensory testing shows unilateral anesthesia. Imaging studies show an infarction as indicated by the arrow and the outline in the figure. Branches of which of the following arteries are most likely implicated in the infarction in this patient?

- A. Left external carotid
- B. Left internal carotid
- C. Left vertebral
- D. Right external carotid
- E. Right internal carotid
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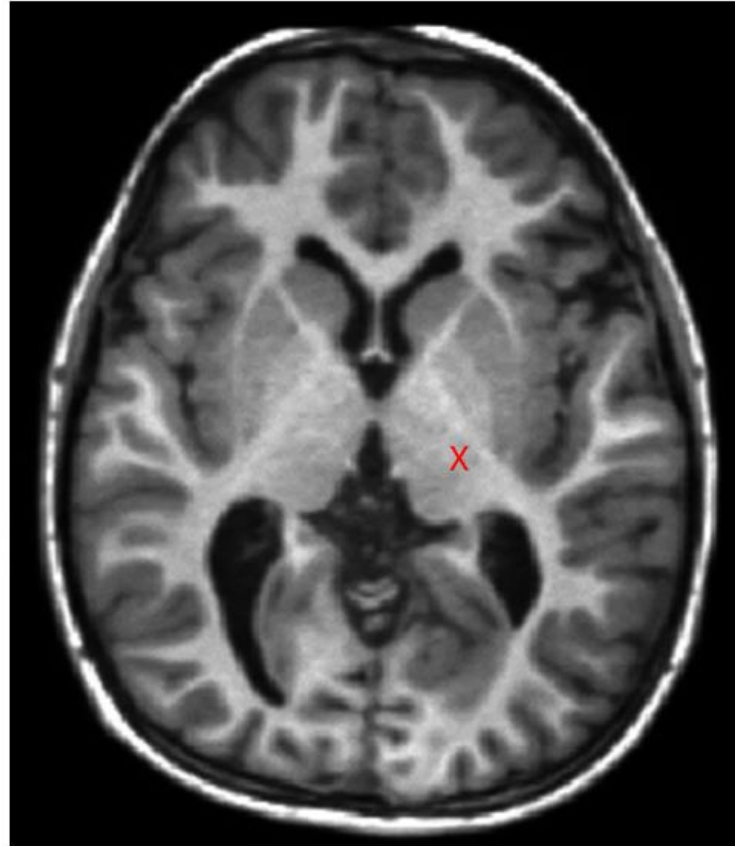
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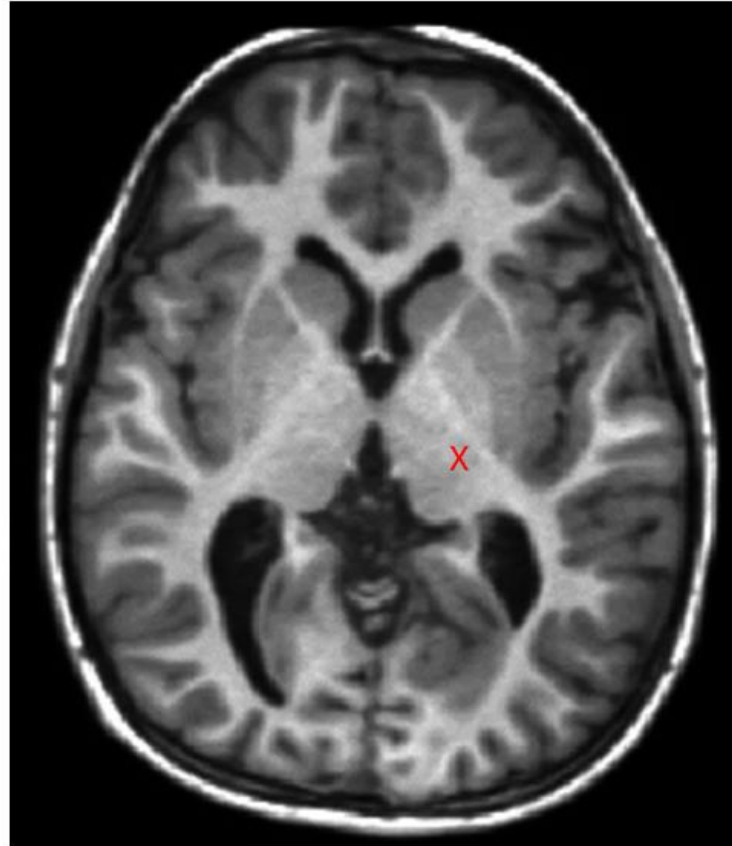
11. A 54-year-old woman comes to the physician because of a 2-day history of sudden-onset numbness affecting the right side of her body. Physical examination shows that strength is normal. Imaging studies show a lesion (indicated by the 'X' in the attached image). Which of the following structures is most likely damaged in this patient?

- A. Caudate
- B. Putamen
- C. Globus pallidus
- D. Optic radiation
- E. Thalamus



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12. A 45-year-old man comes to the physician because of a 3-week history of painful eyes and blurry vision. Neurologic examination shows bilaterally normal strength. MRI of the brain shows an inflammatory demyelinating lesion affecting the right optic nerve. Laboratory studies show elevated antibody titers. Which of the following cell types is most likely damaged by the disease process in this patient?

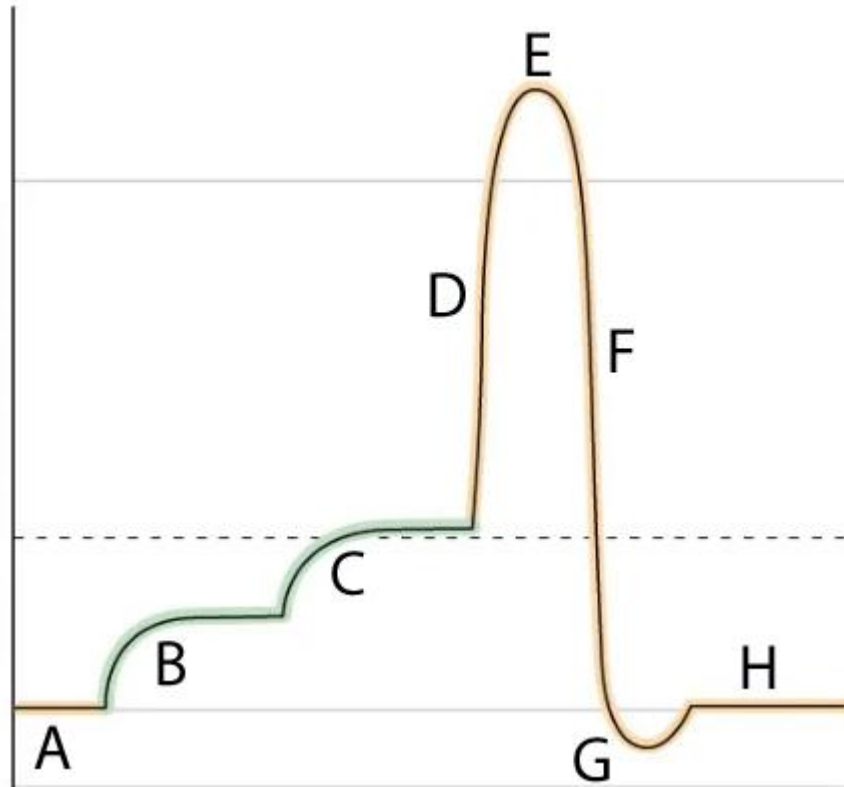
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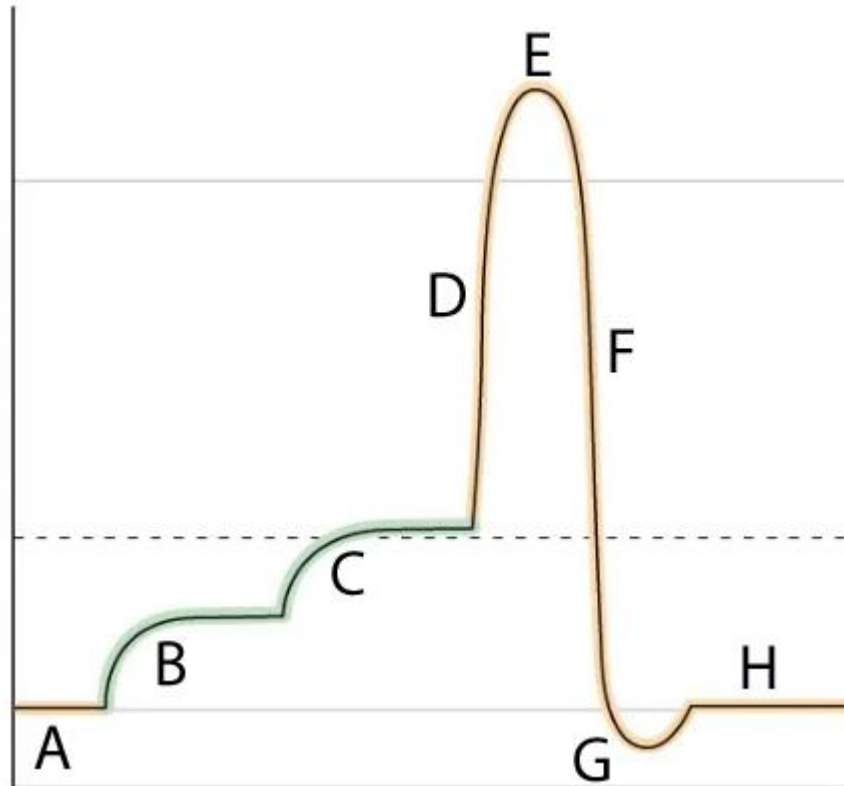
13. Investigators are studying the electrical events in a postsynaptic neuron in response to excitatory input from a presynaptic terminal. Which of the following labeled electrical events (see figure) best represents the opening of ligand-gated sodium channels during temporal summation?

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14. Investigators are studying the propagation of action potentials in axons of cultured mammalian multipolar neurons. With sufficient depolarization of neuronal cell bodies, it is observed that action potentials arise and propagate from the axon hillocks to the axonal terminals. Which of the following terms best applies to the observed pattern of propagation of the action potentials expressed by the cells under study?

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